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# Neural Implicit Quantitative Imaging

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## 1 Introduction

Magnetic Resonance Imaging (MRI) is a well-known method and an indispensable tool in clinical practice. However, the most commonly used protocols are qualitative, i.e. a mixture of different tissue parameters and acquisition properties determines the contrast in the images. In contrast, quantitative MR imaging (qMRI) aims to measure biophysical tissue parameters, for example, longitudinal relaxation times (T1).

Specifically, in cardiovascular MR, T1 mapping has been translated into clinical application and can be used to diagnose various cardiac pathologies [5, 12]. In practice, it is limited by the trade-off between scan time and signal-to-noise ratio, as well as the complexity of the reconstruction problem. Despite the general success of learned regularization for MR image reconstruction [13, 6], the application to high-resolution, 3D quantitative imaging is hindered by the limited availability of training data and the substantial computational and storage demands associated with 3D voxel representations. Consequently, commonly only stacks of 2D maps with reduced through-plane resolution are considered [6, 18].

Neural implicit fields offer an innovative means of representing multidimensional data by directly mapping input coordinates to various quantities, leveraging familiar neural network components [10]. Notably, neural fields can be used for subject-specific optimization without any additional training data, benefiting from the architecture's efficiency, inherent regularization properties, and stochastic optimization. While neural fields have exhibited effectiveness in diverse MR applications [16, 3, 17, 15, 4, 2, 7, 9], they have to our knowledge not been used for qMRI.

Here, we propose an end-to-end approach encompassing motion-corrected reconstruction, coil-sensitivity map estimation, slice-to-volume super-resolution, and quantitative mapping in 3D. We report initial experiments with this novel technique for whole-heart T1 mapping.

## 2 Methods

We recast the well-known quantitative imaging inverse problem of obtaining voxel-wise parameter maps from acquired k-space data  $k$  to

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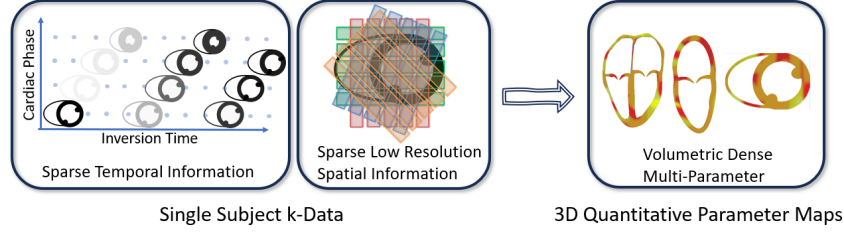


Figure 1: The inverse problem of super-resolved quantitative cardiac imaging. From k-data of thick slices recorded in a few orientations and positions, we aim to reconstruct full 3D T1 maps. Note also the sparsity in inversion time/cardiac phase, as for each slice only a single inversion recovery is acquired.

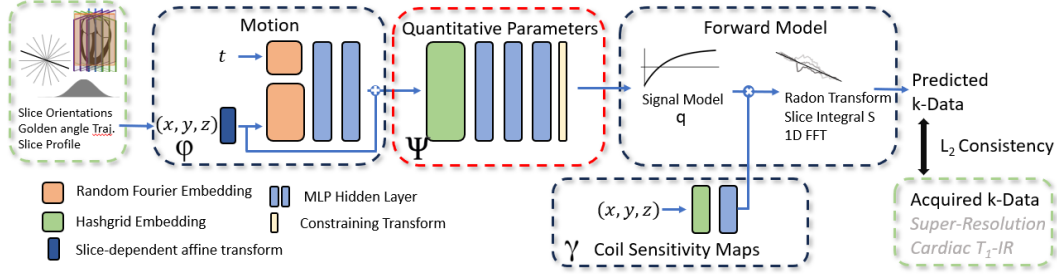


Figure 2: Schematic overview. We model the 3D parameter fields (red) by a coordinate-based network. We include additional fields for motion and the coil sensitivity maps. Our forward model consists of the signal function  $q$ , the integral over the slice profile and a radial Fourier transform (implemented via Radon Transform). All weights are optimized per subject, minimizing the discrepancy between predicted k-data and acquired k-data

$$\min_{\phi, \psi, \gamma} \|\mathbf{k} - \mathbf{FC}_{\gamma}(\mathbf{r})\mathbf{S}q(\psi(\mathbf{r}'))\| \quad (1)$$

$$\mathbf{r}' = \psi \mathbf{r} \quad (2)$$

With signal model  $q$ ,  $\mathbf{S}$  an integration operator over the slice profile, sensitivity map dependent coil-expand operator  $\mathbf{C}$ , and Fourier transform  $\mathbf{F}$ . Here, we introduce a continuous parameter field  $\psi$  mapping from 3D coordinates  $\mathbf{r}$  to quantitative parameters, a motion transform  $\phi$ , and coil sensitivity field  $\gamma$ . We model  $\phi$ ,  $\psi$ , and  $\gamma$  by neural fields. Hence, our proposed architecture (Figure 1) consists of 1) a neural motion field. 2) a coil sensitivity field. 3) the main quantitative parameter field, mapping coordinates to the parameters of interest (e.g. T1). 4) a forward model, applying the signal model (here: cont. acquisition inversion recovery) and obtaining for each mini-batch the estimated k-space signal by applying the sensitivity maps and the radial Fourier transform. We employ the Fourier slice theorem to capitalize on the ability to sample our model in arbitrary orientations, thus avoiding a NUFFT. The forward model includes the slice profile via Monte-Carlo integration.

As in prior works, the motion field comprises a slice-conditioned affine transform, random Fourier embeddings [14] in space (256 features) and time (64 features), and a 2 hidden layer MLP (256 features) [15, 2]. The coil sensitivity module [4] as well as the parameter module use a Hashgrid embedding [11]. For the sensitivity maps, we limit the capacity by using only 3 resolutions in the embedding and a single hidden layer, while the parameter network uses 18 different resolutions (scaling factor 1.3, hash length  $2^{22}$ , and 3 hidden layers (256 features)). We use the output activation of the parameter field to avoid impossible parameter values, i.e. we enforce positivity of T1 by a Softplus.

The trainable parameters of the neural fields are optimized end-to-end, minimizing (1) using AdamW with cycled learning rate, weight decay=1e-3, beta1=0.99, and mini-batches of 8 spokes. Total runtime was 30min (Nvidia A6000). We promote smooth sensitivity maps by including Gaussian

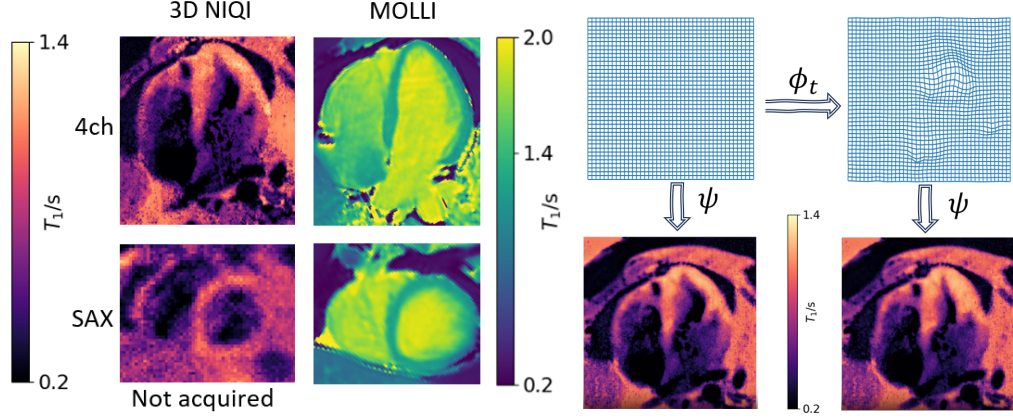


Figure 3: Results for T1 maps (left) compared to a reference MOLLI acquisition and effect of the learned motion field (right).

jitter to the respective input coordinates and volume preservation in the motion module by a Jacobian determinant penalty.

Data was collected from a healthy volunteer with a 3T Siemens Verio with 32-channel cardiac coil (compressed to 8 channels). Three stacks, consisting of five slices each, were acquired in four LAX orientations [8, 1]. After a slice-selective inversion pulse, data was continuously acquired using a golden angle radial scheme with spatial resolution  $1.3 \times 1.3 \times 4.0$  mm<sup>3</sup>. Total scan time 3min in 12 breath-holds. For comparison, a 3(3)3(3)5-MOLLI was acquired in 4ch and SAX orientation.

### 3 Results

Initial results demonstrate the ability to reconstruct quantitative 3D T1 maps from multislice 2D acquisitions (Figure 2). These maps can be visualized in explicitly acquired orientations (4ch-view) and non-acquired orientations (SAX) by evaluation of  $\psi$  (Figure 3), illustrating successful 3D representation (Figure 4). Additionally, Figure 5 illustrates the effect of the cardiac motion field on a single slice.

### 4 Discussion

Modeling the parameter maps using neural fields shows great potential in qMRI. Our current approach already replaces hand-crafted multi-step pipelines with a single model-based end-to-end optimization while not relying on any training data. Even without any regularization on the parameter maps, we obtain in our initial results parameter maps in orientations not explicitly acquired in the challenging problem of full-heart T1 mapping with 100% scan efficiency. The possibility of evaluating the fields in arbitrary orientations and resolutions allows for many regularisation approaches, which are known to improve fidelity, and thus will be further investigated.

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